

HANITH TUMMALAPALLI

Data Engineer | Analytics Engineer

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Summary

Data Engineer with 3+ years of experience building batch and real-time streaming pipelines across healthcare and data center infrastructure domains. Hands-on with Kafka, Spark Structured Streaming, Databricks (Delta Lake, DLT), Snowflake, dbt, SSIS, Azure Data Factory, and Power BI; built production RAG/LLM systems and ML pipelines on 230K+ patient records. MS in Data Science (4.0 GPA); Microsoft and Snowflake certified.

Technical Skills

Languages: Python, SQL (CTEs, window functions, query optimization), PySpark, DAX

Streaming & Big Data: Apache Kafka, Confluent Cloud, Spark Structured Streaming, Databricks, Delta Lake, Lakeflow Declarative Pipelines (Delta Live Tables), Medallion Architecture, Unity Catalog, Auto CDC, schema evolution

ETL & Orchestration: SSIS, Azure Data Factory, dbt, incremental ETL, data validation, REST API ingestion

Data Warehousing: Snowflake, Azure Synapse, SQL Server, PostgreSQL, MySQL, dimensional modeling, star schema

BI & Analytics: Power BI (DAX, data modeling), Tableau, Microsoft Fabric, Excel (Power Query, Power Pivot)

AI/ML & Tools: RAG, LLMs (Ollama, Gemma), scikit-learn, model evaluation (AUC, precision/recall), Power Apps, Django, Git

Experience

Advanced Medical Transport — Data Science Intern, Peoria, IL **May 2026 – Present**

- Designed and deployed an on-premises Retrieval-Augmented Generation (RAG) system using Ollama and Gemma LLM for secure semantic search across internal documentation, reducing document lookup time by 60%.
- Built SQL Server ETL pipelines in SSIS integrating CAD (computer-aided dispatch), billing, and operational EMS data, cutting manual processing by 40% and improving reporting accuracy.
- Automated HIPAA-compliant exception reporting and data validation workflows, saving 8+ hours per week and strengthening compliance and data integrity.
- Developed Power Apps applications for internal operations and partnered with EMS operational teams to gather requirements, analyze business processes, and deliver workflow improvements.

Bradley University — Research Assistant (Data Science), Peoria, IL **Aug 2025 – May 2026**

- Built end-to-end ML pipelines on 230,000+ patient records in collaboration with OSF HealthCare to predict heart disease and diabetes, improving prediction accuracy by 8%.
- Engineered data cleaning and feature-engineering workflows in Python and SQL, delivering analytics-ready datasets and automating reproducible training/validation runs evaluated on AUC, precision, recall, and F1.

Praval Infotech — Data Engineer / ETL Developer, Hyderabad, India **Nov 2023 – Jan 2025**

Client: Digital Realty — Data Center & Cloud Infrastructure Analytics

- Developed and maintained SSIS packages automating data loads into SQL Server, improving ETL reliability and reducing job failures by 40%.
- Optimized SQL queries and data models, improving data refresh speed and Power BI dashboard load times by 30% for operational and executive reporting.
- Delivered analytics dashboards on SQL Server and Microsoft Fabric by gathering requirements and defining KPIs with stakeholders for supply chain and operational insights.

Vertocity — Data Analyst, Hyderabad, India **Jul 2022 – Oct 2023**

- Developed a Django-based ERP system centralizing student progress and staff activity tracking, and analyzed academic/operational datasets to surface performance trends.
- Built Excel dashboards with Power Query and Power Pivot, reducing recurring reporting time by 30%.

Projects

Real-Time Healthcare Streaming Pipeline — Kafka, Confluent Cloud, Databricks, Delta Lake, Python

- Built an event-driven streaming pipeline ingesting adverse drug event data from the OpenFDA REST API via a Python Kafka producer publishing to Confluent Cloud, processed in Databricks with Spark Structured Streaming.
- Implemented Medallion Architecture (Bronze/Silver/Gold) with Lakeflow Declarative Pipelines (DLT): append-only raw ingestion with schema evolution, flattening nested JSON into normalized streaming Delta tables, and analytics-ready Gold tables using Auto CDC for incremental merges, deduplication, and Unity Catalog governance.

Healthcare Data Pipeline — Azure Data Factory, Synapse, Power BI

- Built ADF pipelines migrating patient and clinical data from on-prem SQL Server to Azure Synapse with governance and data quality checks; delivered Power BI dashboards for clinical decision-making.

Financial Data Warehouse — dbt, Snowflake, Power BI

- Modeled financial datasets into star schema tables with dbt transformations in Snowflake; built Power BI dashboards with drill-through analysis for financial planning.

DataMate — AI-Powered Natural Language to SQL Engine (Python, Django)

- Built an LLM-driven query engine translating natural language to SQL across SQL Server, PostgreSQL, and MySQL backends, reducing query turnaround by 70%.

Education

Bradley University — MS, Data Science (GPA: 4.0), Peoria, IL **Jan 2025 – Dec 2026 (expected)**

Coursework: Big Data Analytics, Statistical Modeling, Machine Learning, Cloud Data Engineering (Azure), Tableau

Certifications

Microsoft Certified: Power BI Data Analyst Associate — Certificate | Snowflake Certified — Certificate